

Olevel economics

Chp:- Demand and Supply

Topic:- Extreme cases of PED

A stylized handwritten signature in black ink, featuring a large loop and a horizontal stroke.

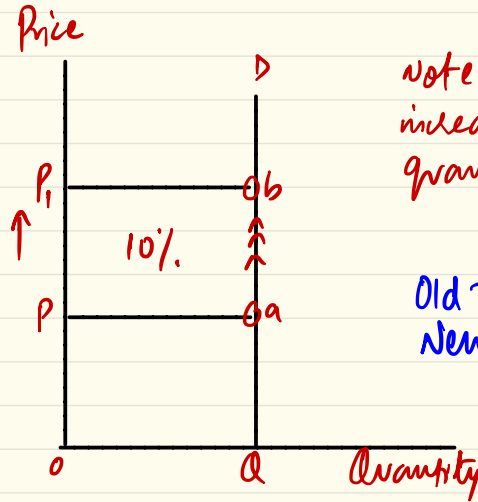
Zarale
Mushraf

Extreme cases of Price Elasticity of demand

$$PED = 0$$

- No responsiveness
- No substitutes
- Life saving drugs
- * Perfectly inelastic demand

Price $\uparrow$ QD - TR $\uparrow$



note:- A 10% rise in price will increase total revenue by 10% as quantity remains unchanged.

Old TR/TE:- $OPaQ$
New TR/TE:- OP_1bQ

Total revenue increased

Note:- In case of perfectly inelastic demand, as there are no substitutes available, a producer will increase the price, which leaves quantity demanded unchanged, causing a rise in total revenue.

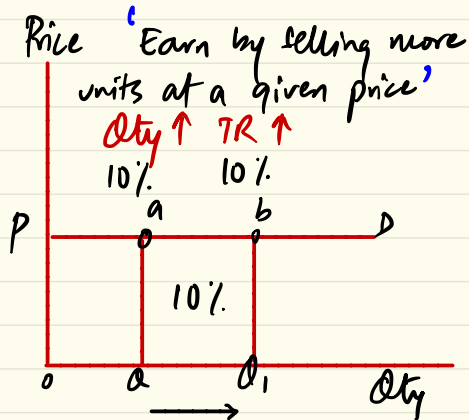
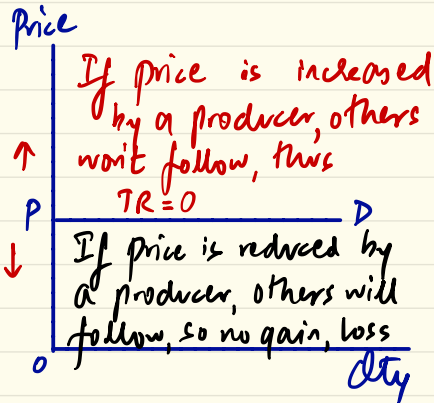
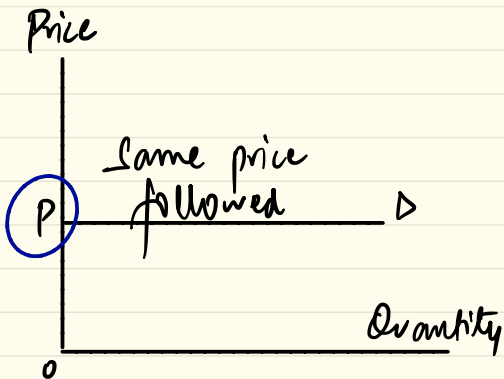
$$PED = \infty$$

→ perfectly elastic demand

☆ perfect substitutes
 ☆ perfect competition
 ☆ perfect knowledge

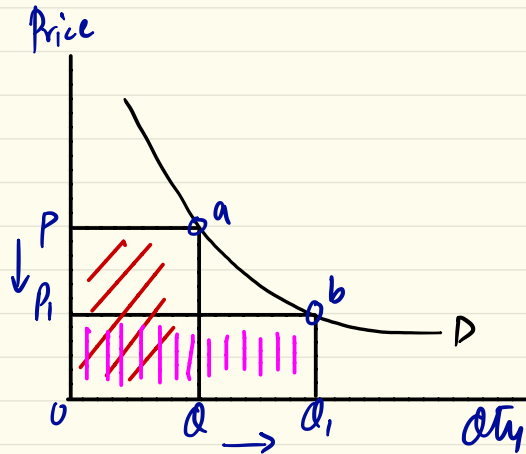
- 1000s of producers
- same / standardized / homogeneous products
- same price
- price taker
- No barriers to entry or exit

old TR :- $OPad$
 new TR :- $OPbd$, $TR \uparrow$



Note:- In case of perfectly elastic demand, there are 1000s of producers, producing and selling same product, and the market share is so small that they can't influence the price level, thus known as price taker.

$PED = 1 \rightarrow$ Unitary :- A change in price brings an equal percentage change in quantity demand. For e.g. a rise in price by 10% will contract the quantity by 10%. Keeping total revenue and total expenditure unchanged.



Old $TR = OPaQ$
New $TR = OP_1bQ_1$

Equal area
 TR/TE :- unchanged

e.g.

Price	Qty
10	5
5	10

$$TR = P \times Q$$

So

So

TR remains unchanged

Questions (H.W #6)

Marks:- 18

- Q1 Using a diagram, explain the concept of perfectly inelastic demand? 6
- Q2 Using a diagram, explain the concept of perfectly elastic demand? 6
- Q3 Using a diagram, explain the concept of Unitary demand? 6



Define each concept, draw the diagram, explain the diagram